

Reducing 30-Day Readmissions: A Healthcare Analytics Case Study

4 October 2026 · Chigozie Nkwopara

1. Organisation and dataset context

Meridian Health Network is a fictional healthcare organisation created for this case study. It represents a network of 130 acute hospitals across the United States.

The analysis uses the [Diabetes 130-US Hospitals dataset](#) by John Clore, Krzysztof Cios, Jon DeShazo and Beata Strack, published in the UCI Machine Learning Repository (DOI 10.24432/C5230J) under a CC BY 4.0 licence. Full attribution is in section 13. It contains 101,766 hospital encounters involving 71,518 patients with diabetes across 130 US hospitals between 1999 and 2008. Each record is one hospital stay of 1 to 14 days.

For the case study, these records are treated as Meridian's historical patient data to simulate a real healthcare analytics project. The risk model is a retrospective analytical prototype for portfolio demonstration. It has not been clinically validated or approved for patient-care decisions.

Context	Detail
Organisation	Meridian Health Network, fictional
Setting	130 US acute hospitals
Patient population	Patients with diabetes
Raw data	101,766 hospital encounters
Patients	71,518
Period	1999 to 2008
Business focus	30-day readmission
Data source	UCI Diabetes 130-US Hospitals dataset
Funding context	Private insurance, Medicare and Medicaid. Payers penalise high 30-day readmission rates

2. Stakeholder discovery

This section is a simulated discovery artefact. No real interviews took place. It shows how the problem would be scoped in a real healthcare analytics engagement.

To define the analytics problem, I designed a discovery questionnaire for four roles involved in readmission management: the Director of Clinical Operations, discharge planning leads, the diabetes clinical lead and the finance director. The aim was to find the operational questions the available data could realistically answer.

Stakeholder question	Purpose	Asked of
What problems create the greatest pressure on inpatient capacity?	Identify the business problem	Director of Clinical Operations

Stakeholder question	Purpose	Asked of
Which patient outcomes cause the greatest operational or financial concern?	Establish priorities	Director of Clinical Operations, finance director
At what point in the patient pathway could your team intervene?	Find actionable decision points	Discharge planning leads, diabetes clinical lead
How is readmission risk identified today?	Understand the current process	Discharge planning leads
Which patients receive follow-up after discharge today?	Identify targeting gaps	Discharge planning leads
What limits the number of patients your team can follow up?	Establish capacity constraints	Discharge planning leads, finance director
Which measures would show that an intervention worked?	Define success measures	All four roles
What information would you need before changing the discharge process?	Define analytical requirements	Director of Clinical Operations, diabetes clinical lead

The discovery is not reported as invented responses. Section 4 translates it into requirements, which gives a clear chain: stakeholder needs → business problem → analytical questions → data requirements → analysis → recommendation.

3. Business problem and objectives

The case scenario assumes that senior management is experiencing pressure from unplanned readmissions, bed demand and financial exposure from payer penalties. In the data, about one in eleven diabetic patients returns to a Meridian hospital within 30 days of discharge.

The discovery framework in section 2 was used to structure where analytics could support operational decisions. Section 4 translates the scenario into decision and analytics requirements. Together they narrow the scenario to one question: which patients have the greatest risk of 30-day readmission, what factors identify them, and how should limited follow-up capacity be targeted?

Objectives. The analysis answers four questions:

1. How often are patients readmitted, and how long do they stay?
2. Which patient, admission and discharge factors link to readmission?
3. Are standard care checks, such as HbA1c testing, done consistently?
4. Can a simple model flag high-risk patients before they leave hospital?

4. Stakeholders, decisions and analytics requirements

Each discovery theme becomes one analytics requirement with a success measure, traced to the section that answers it.

Stakeholder	Identified challenge	Decision required	Analytics requirement	Success measure	Answered in
Director of Clinical Operations	Readmissions consume beds and staff time	Where should resources go?	Identify the main readmission drivers	Readmission rate and effect size by risk factor	Sections 7 and 8
Discharge planning leads	Follow-up capacity is limited	Who receives follow-up?	Patient-level risk score with capacity scenarios	Recall, precision and lift at each capacity level	Section 8
Diabetes clinical lead	Possible variation in diabetes monitoring	Should care processes change?	Analyse HbA1c testing rates and their link to readmission	Testing rate and its variation across groups	Section 7
Finance director	Any intervention needs funding	Is targeted follow-up financially justified?	Estimate the intervention population and potential benefit	Net benefit, cost per prevented readmission and break-even prevention rate	Section 9

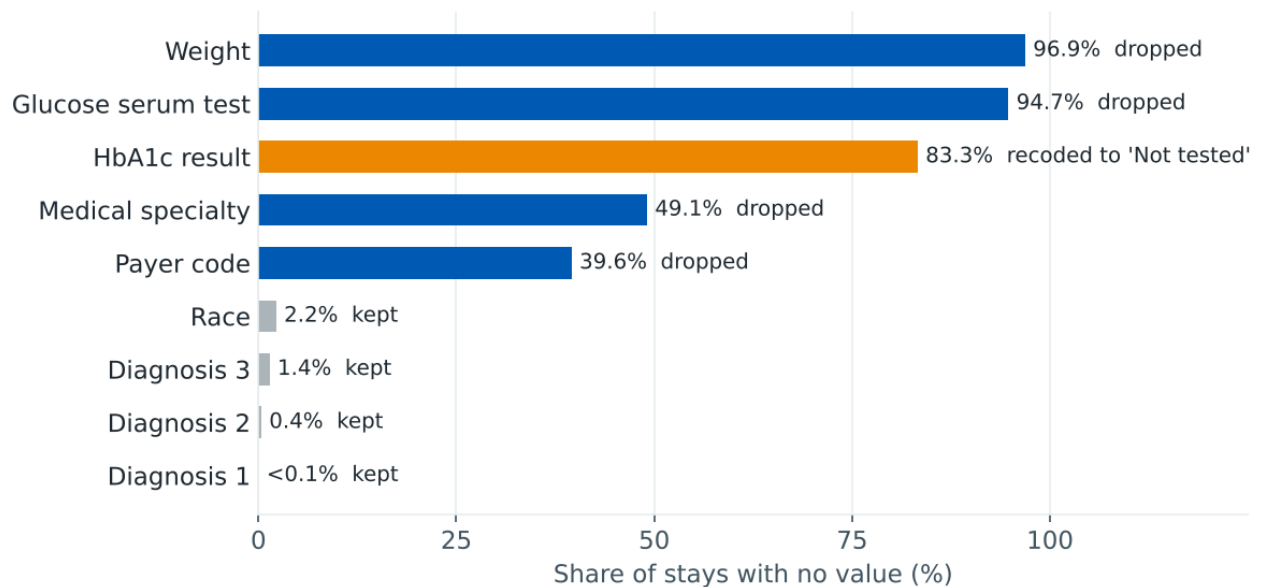
5. Data quality assessment

The raw extract had ten problems that would have distorted the results if left in place.

Challenge	Scale	Risk if ignored	Fix
Missing values coded as "?"	Every text column	Gaps read as real values	Read "?" as missing on load
Weight almost empty	96.9% missing	No usable signal	Dropped
Glucose serum test almost empty	94.7% missing	No usable signal	Dropped
Medical specialty half empty	49.1% missing	Biased comparisons by specialty	Dropped
Payer code patchy	39.6% missing	No link to readmission question	Dropped
HbA1c result blank for most stays	83.3% blank	Blank looks like missing data, but means "not tested"	Recoded to "Not tested" and kept as a care quality measure
Patients who died in hospital or went to hospice	2,423 stays	Inflates the "not readmitted" group	Removed
Repeat patients	30,248 extra stays from the same people	Correlated observations break the independence that tests assume, and one patient could sit in both training and test samples	Kept one index stay per patient for this exploratory analysis
Coded fields	Admission, discharge and diagnosis stored as numbers and ICD-9 codes	Results unreadable for stakeholders	Decoded with the mapping file and grouped
Invalid gender	3 records	Minor noise	Removed

Four fields were too empty to use

Missing values by field in the raw extract, 101,766 stays



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. Analysis: C. Nkwopara.

6. Data preparation

Cleaning in Python (pandas) reduced the data to 69,987 patients, one stay each, with 58 columns. The final 30-day readmission rate is 9.0%.

Step	Rows after	Patients after	Change
0. Load raw data, "?" read as missing	101,766	71,518	Starting point
1. Drop weight, payer code, medical specialty, glucose serum	101,766	71,518	4 columns removed
2. Remove invalid gender	101,763	71,515	-3 rows
3. Remove deaths and hospice discharges	99,340	69,987	-2,423 rows
4. Keep first stay per patient	69,987	69,987	-29,353 rows
5. Decode codes, group diagnoses, add derived fields	69,987	69,987	12 fields added

Derived fields added: 30-day readmission flag, age band and midpoint, HbA1c tested flag, total prior visits, number of medication changes, grouped admission source, grouped discharge destination and grouped primary diagnosis.

Key fields in the analysis-ready dataset:

Field	Type	Description
readmitted_30	Yes or no	Outcome: returned within 30 days of discharge
time_in_hospital	Days (1 to 14)	Length of stay
age_band	10-year bands	Patient age group
gender, race	Category	Demographics

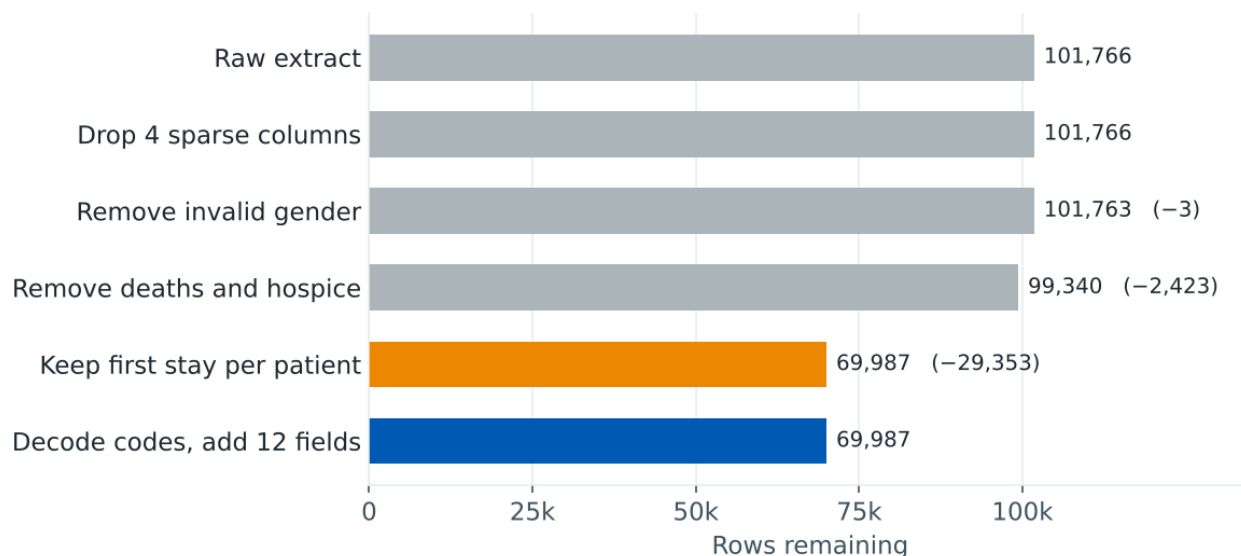
Field	Type	Description
admission_type	Category	Emergency, urgent, elective
admission_source_group	Category	Emergency room, referral, transfer
discharge_group	Category	Home, home with home health, nursing facility, other inpatient or rehab
primary_diagnosis_group	Category	Main condition, grouped from ICD-9 codes
number_inpatient	Count	Inpatient stays in the year before admission
number_emergency	Count	Emergency visits in the year before admission
num_lab_procedures	Count	Lab tests during the stay
num_medications	Count	Distinct medications given
number_diagnoses	Count	Diagnoses recorded
a1c_tested	Yes or no	HbA1c test done during the stay
med_changes	Count	Diabetes medications increased or decreased
insulin, diabetesMed	Category	Insulin change, and whether any diabetes drug was prescribed

The readmission rate falls from 11.2% in the raw data to 9.0% after cleaning. Patients with more than one stay readmit at 19.8%, against 4.3% for one-stay patients. Recurrent admissions are part of the operational problem, not noise.

For this exploratory patient-level analysis, one index encounter per patient was retained to avoid correlated observations and patient leakage between training and test samples. This simplifies statistical testing but removes information about recurrent admission patterns. A production analysis should retain longitudinal encounters and use patient-grouped validation or repeated-measures methods.

Keeping one stay per patient removed 29,353 rows

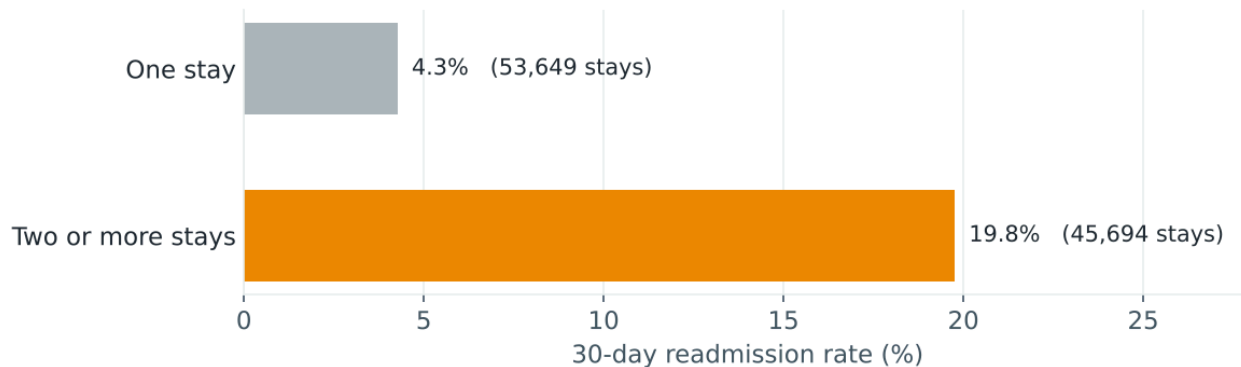
Rows after each cleaning step. Final dataset: 69,987 patients, one stay each



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. Analysis: C. Nkwopara.

Repeat patients readmit at over four times the rate

Recurrent stays carry real signal. One index stay per patient was kept to avoid correlated data



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. Analysis: C. Nkwopara.

Code: 01_clean_data.py builds the clean dataset and cleaning_log.csv. 02_visualise_cleaning.py reads both files and draws these charts, so a change to any cleaning step flows through to the charts.

7. Analysis

7.1 Approach

Each objective from section 3 maps to a set of measures from the top 20 healthcare list.

Objective	Measures	What the answer looks like
1. How often are patients readmitted, and how long do they stay?	Percentage share, mean, median, standard deviation, IQR, percentiles	Readmission rate with a confidence interval, length of stay profile

Objective	Measures	What the answer looks like
2. Which factors link to readmission?	Chi-square, confidence interval, t-test, Mann-Whitney U, correlation, effect size	Readmission rate by age, admission source, discharge destination, diagnosis and prior visits
3. Are care checks done consistently?	Percentage share, chi-square, z-score	HbA1c testing rate by group, and its link to readmission
4. Can a model flag high-risk patients?	AUC-ROC with confidence interval, cross-validation, PR-AUC, Brier score, calibration, confusion matrix, precision, recall, lift	Logistic regression risk score with capacity scenarios and subgroup checks

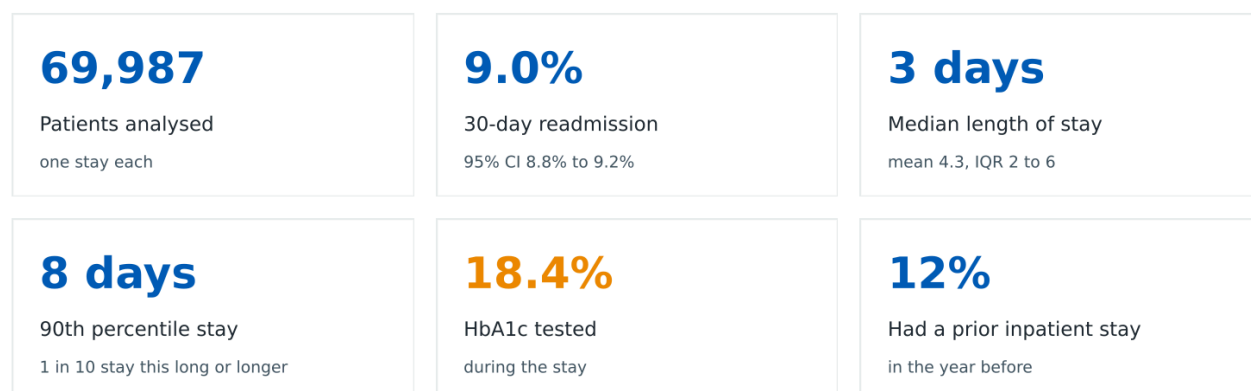
Tools: Python (pandas, SciPy, scikit-learn) for analysis, Power BI for the dashboard, GitHub for the code.

7.2 Descriptive findings

Readmission sits at 9.0% (95% CI 8.8% to 9.2%), but varies from 6.9% to 29.3% across patient groups. Prior hospital use, discharge destination and length of stay show the widest gaps.

About 1 in 11 diabetic patients returns within 30 days

Headline measures for Meridian Health Network, diabetic inpatients

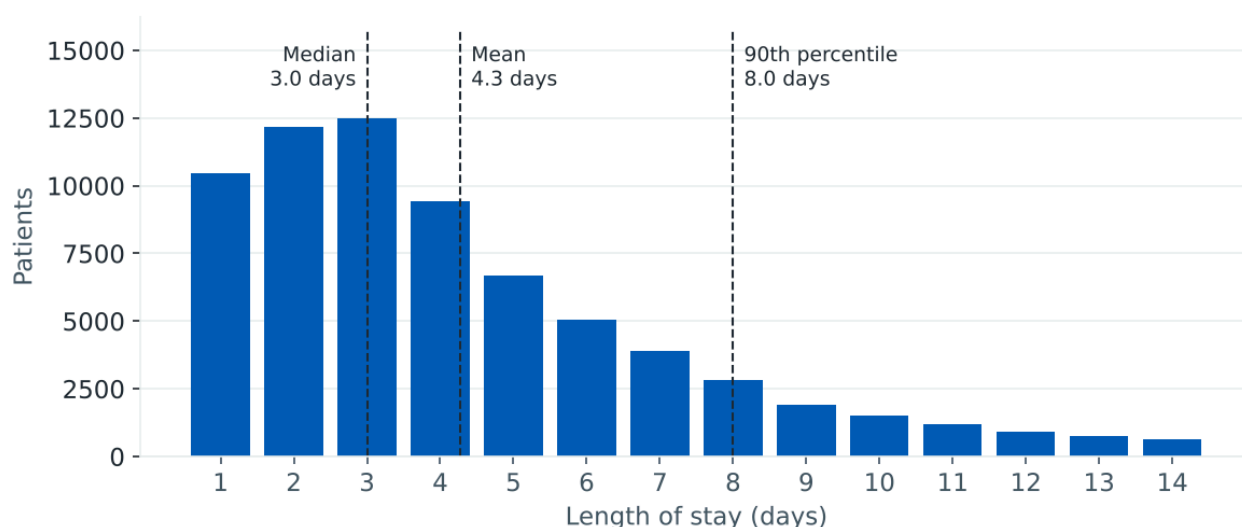


Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Length of stay is right-skewed: median 3 days, mean 4.3 days, 90th percentile 8 days. Reports should quote the median.

Most stays are short, with a long tail (skew 1.18)

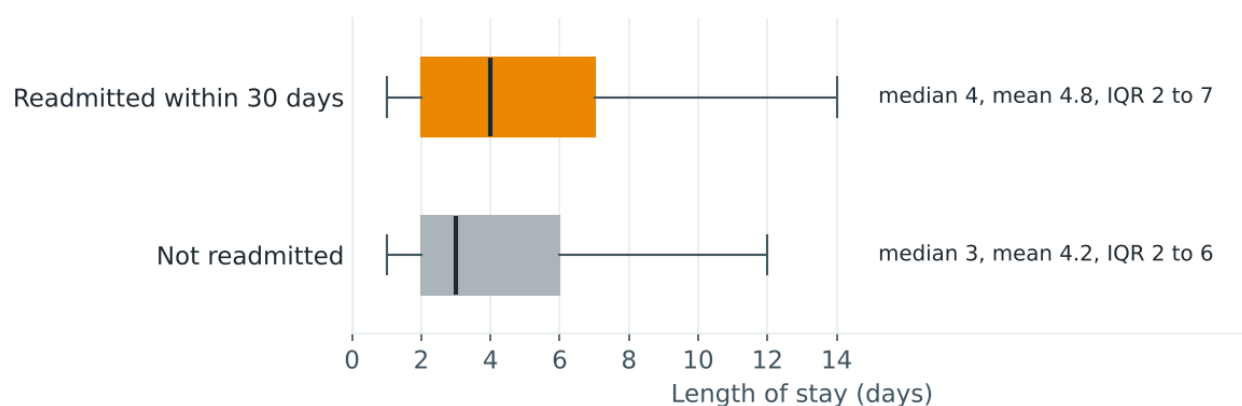
Length of stay for 69,987 patients. Mean sits above the median because long stays pull it up



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Readmitted patients had longer first stays: median 4 days against 3

First-stay length by outcome. Box = middle 50%, line = median, whiskers = range within $1.5 \times \text{IQR}$



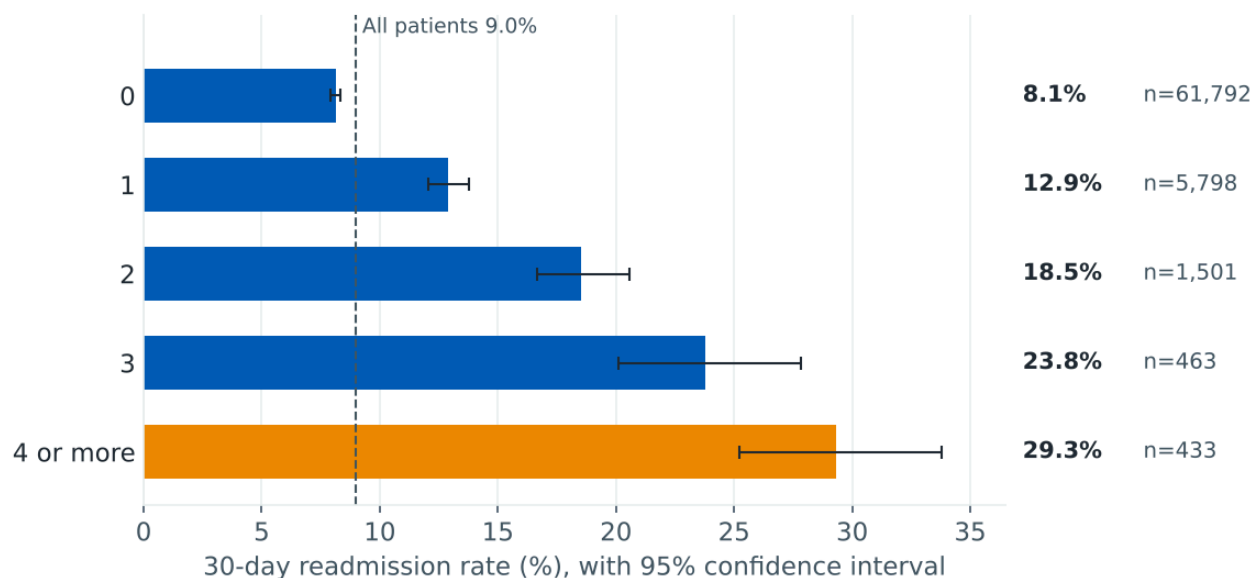
Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Readmitted patients had longer first stays, a median of 4 days against 3.

Prior inpatient stays show the steepest gradient. Readmission rises from 8.1% with none to 29.3% with four or more. Patients discharged to rehab or other inpatient care readmit at 19.4%, nearly three times the 6.9% rate for those sent home.

Readmission risk rises with every prior hospital stay

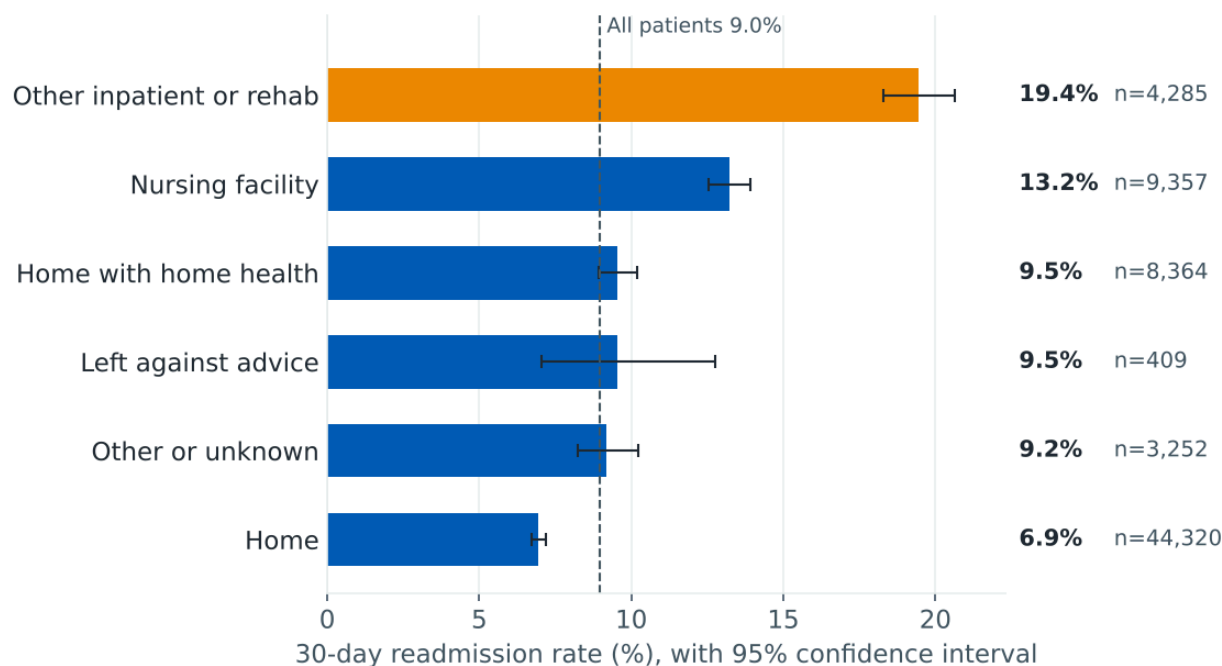
30-day readmission rate by inpatient stays in the year before admission



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Patients sent to rehab or other inpatient care readmit most

30-day readmission rate by discharge destination. Groups under 100 patients hidden

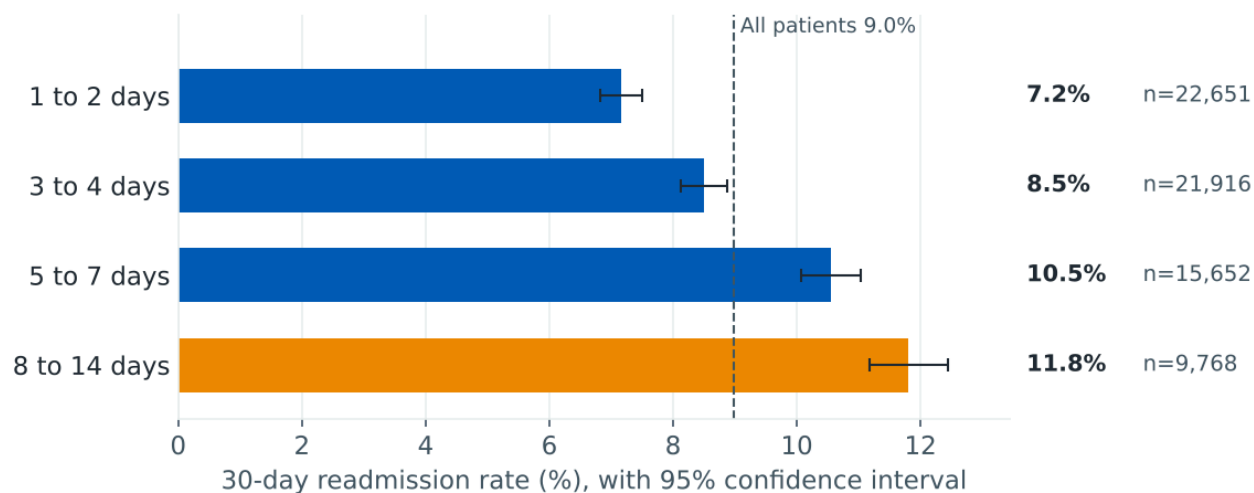


Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Longer first stays carry more risk: 7.2% for 1 to 2 days, 11.8% for 8 to 14 days.

Longer first stays carry higher readmission risk

30-day readmission rate by length of the first stay

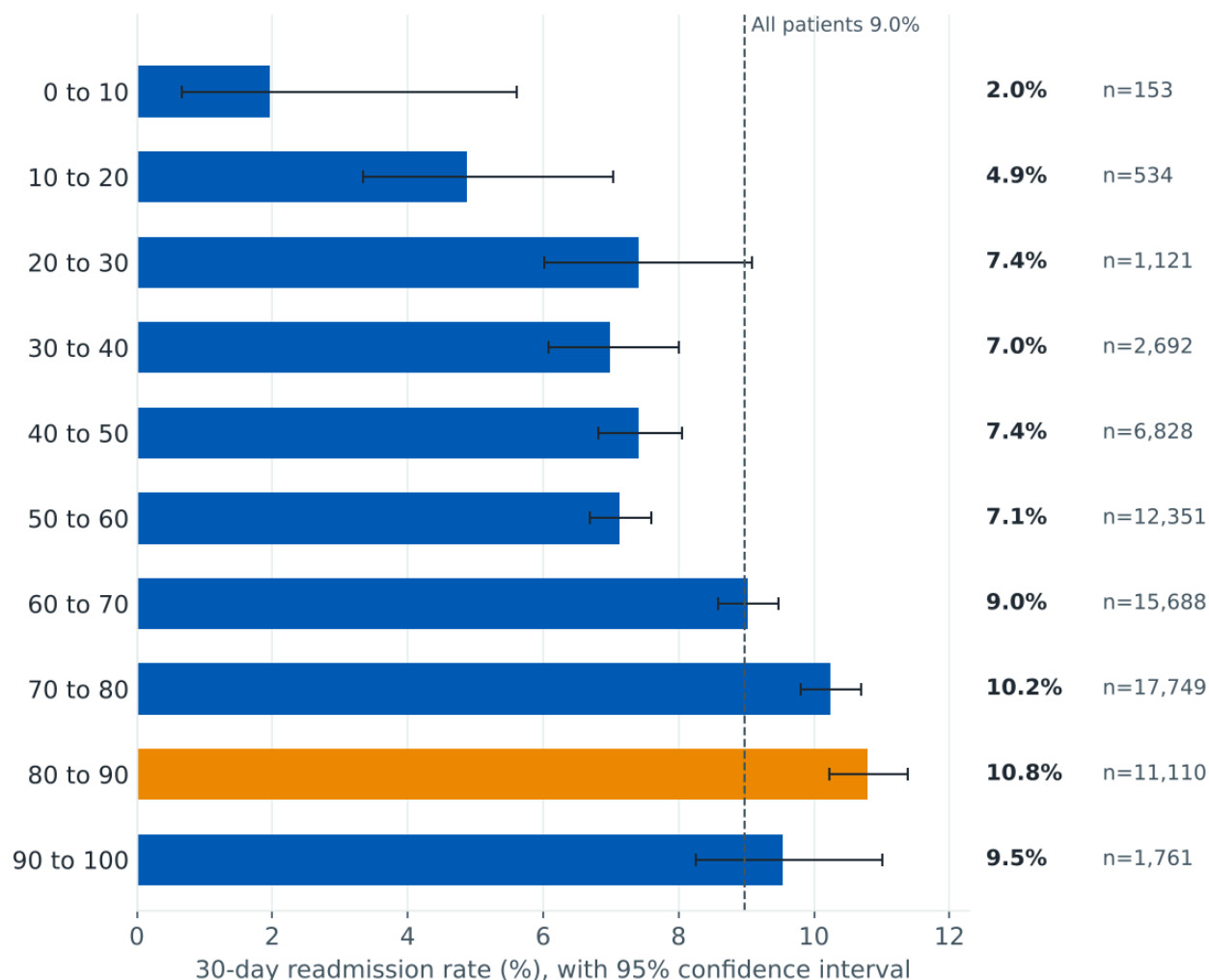


Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Age and diagnosis show smaller differences. Patients aged 70 to 90 readmit at 10% to 11%, against about 7% for those aged 30 to 60. Injury admissions readmit at 10.8% and respiratory at 7.3%.

Readmission climbs gently with age after 50

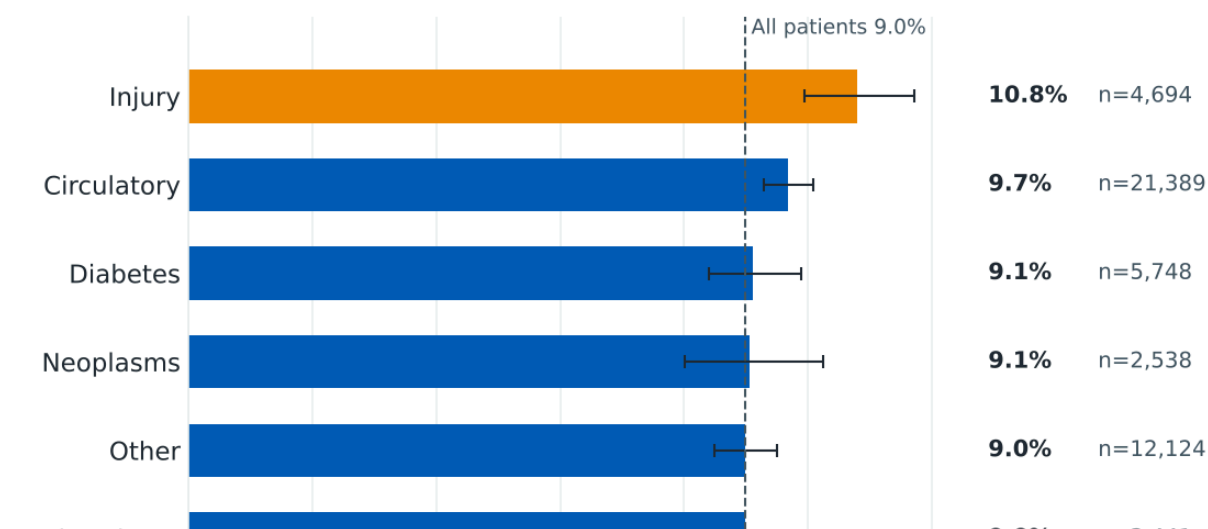
30-day readmission rate by age band. Bands under 100 patients hidden

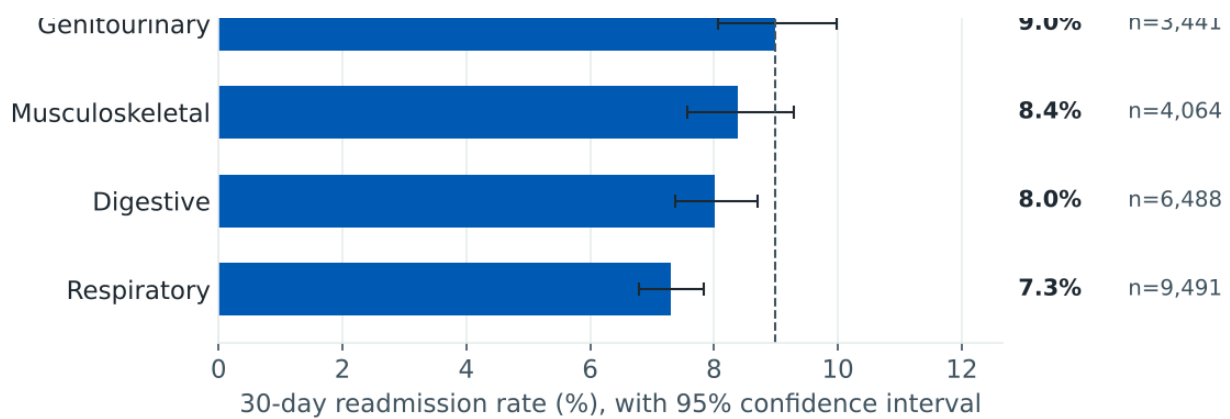


Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Injury and circulatory admissions readmit most often

30-day readmission rate by primary diagnosis group (ICD-9)





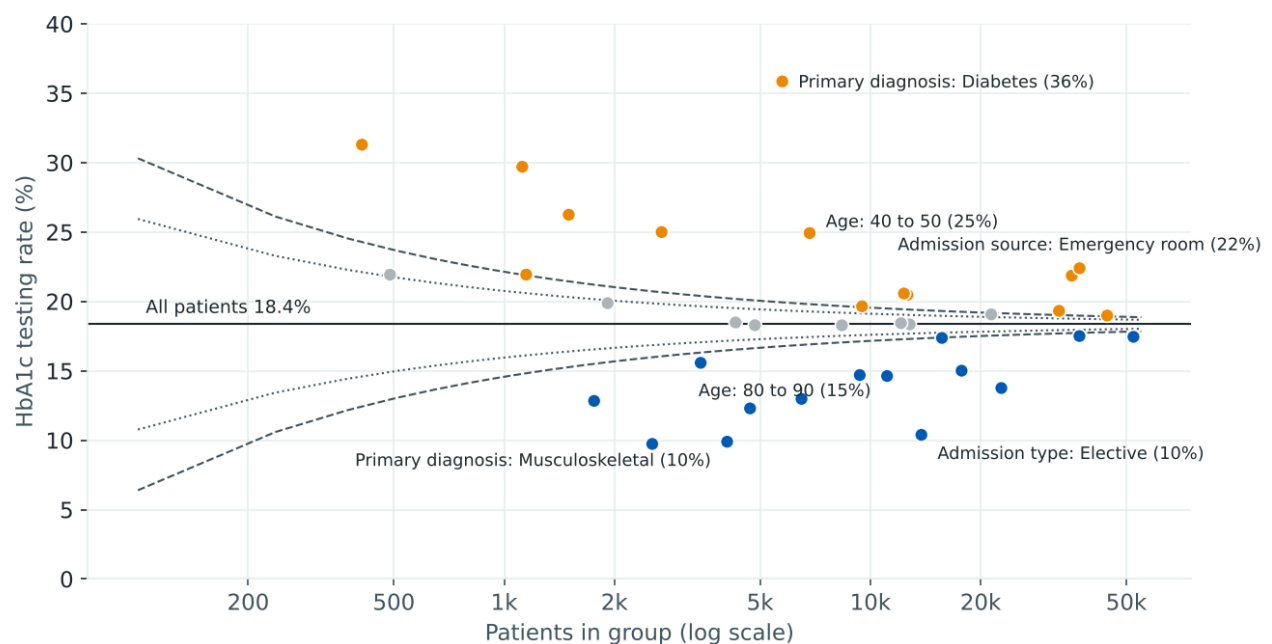
Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Only 18.4% of patients had an HbA1c test during their stay. Tested patients readmitted at 8.4%, against 9.1% for untested patients. Of those tested, 49% had a result above 8%, a sign of poorly controlled diabetes.

Testing varies widely by group. The funnel plot compares each group's rate with the overall 18.4%. Points outside the dashed lines differ by more than chance would explain (z-score beyond ± 3.09).

Elective and older patients are tested least, diabetes admissions most

Funnel plot of HbA1c testing by subgroup. Lines: 95% and 99.8% limits. Orange above, blue below 99.8%



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Groups under 100 hidden. Analysis: C. Nkwopara.

Group	Patients	Testing rate	z-score
Primary diagnosis: diabetes	5,748	35.9%	+34.3
Aged 40 to 50	6,828	24.9%	+14.0
Admitted through the emergency room	37,271	22.4%	+20.1
All patients	69,987	18.4%	0
Aged 70 to 80	17,749	15.0%	-11.5
Aged 80 to 90	11,110	14.6%	-10.2

Group	Patients	Testing rate	z-score
Admitted by referral	22,792	13.8%	-17.9
Elective admission	13,786	10.4%	-24.2
Primary diagnosis: musculoskeletal	4,064	9.9%	-13.9

Testing appears linked to the reason for admission rather than to diabetes itself. Every patient in the data has diabetes, yet those admitted electively or for a musculoskeletal problem are tested at about 10%. Older patients, who carry higher readmission risk, are tested least among adults. This is the variation the diabetes clinical lead's success measure asks for (section 4).

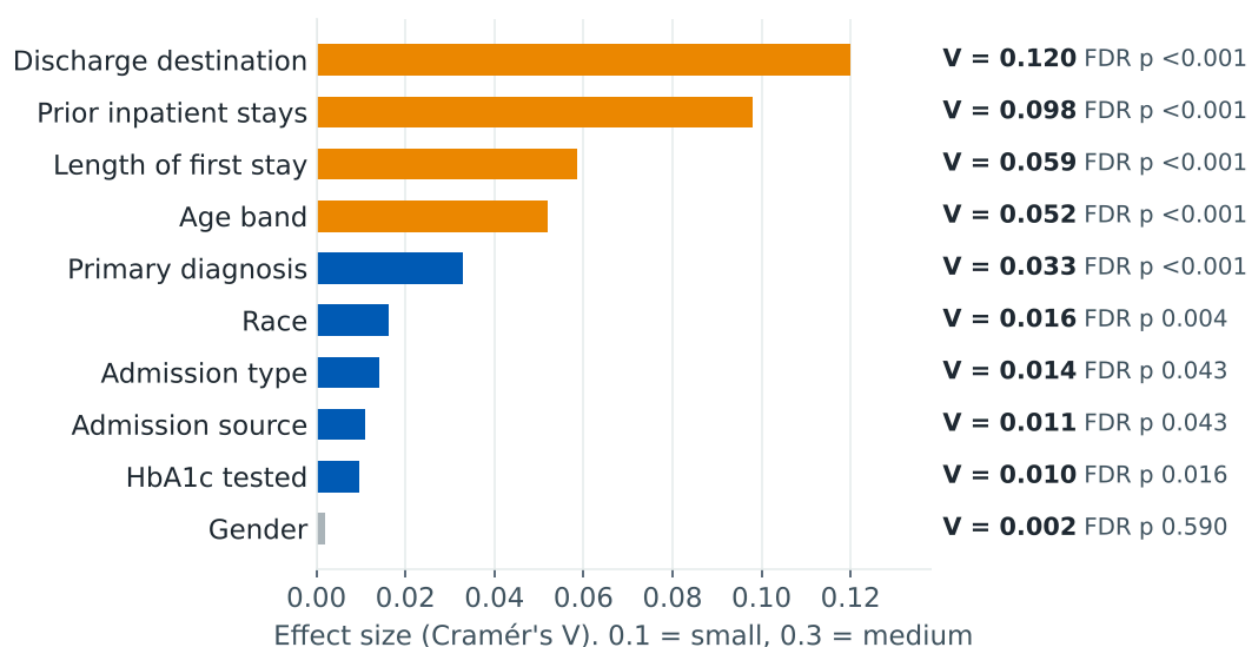
Code: 03_descriptive_stats.py produces these charts plus descriptive_summary.csv and readmission_by_group.csv.

7.3 Statistical tests

Nine of ten factors link to readmission. Only gender does not. Because the large sample makes small differences statistically detectable, interpretation prioritised effect sizes and confidence intervals. The tests were treated as exploratory, with Benjamini-Hochberg false discovery rate (FDR) adjustment applied across the ten categorical comparisons. All nine factors stay significant after adjustment.

Prior stays and discharge destination have the largest effects

Exploratory chi-square tests, FDR-adjusted. Orange: V of 0.05 or more. Grey: not significant



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 69,987 patients, first stay only. Analysis: C. Nkwopara.

Discharge destination (Cramér's V 0.12) and prior inpatient stays (V 0.10) have the largest effects. Length of stay and age follow at about 0.05. HbA1c testing is significant (p = 0.011) but its effect is negligible (V 0.010).

Numeric comparisons, readmitted against not readmitted:

Measure	Median, readmitted	Median, not	Mean, readmitted	Mean, not	Mann-Whitney p	Cohen's d
Inpatient stays, prior year	0	0	0.37	0.16	<0.001	0.35
Length of stay (days)	4	3	4.8	4.2	<0.001	0.20
Diagnoses recorded	8	8	7.5	7.2	<0.001	0.16
Medications	15	14	16.6	15.6	<0.001	0.13
Lab procedures	46	44	44.9	42.7	<0.001	0.11
Emergency visits, prior year	0	0	0.15	0.10	<0.001	0.10

All gaps are real but small. Prior inpatient stays come closest to a medium effect ($d = 0.35$). Mann-Whitney U was used because these counts are skewed. Welch t-tests give the same conclusions.

Spearman correlations confirm the picture. No numeric input correlates with readmission above 0.08. Inputs overlap only moderately, with the strongest pair at 0.46, so collinearity does not distort the model.

Input	Correlation with 30-day readmission	Strongest link to another input
Prior inpatient stays	0.08	Prior emergency visits (0.14)
Length of stay	0.06	Medications (0.46)
Age	0.05	Diagnoses recorded (0.21)
Medications	0.04	Length of stay (0.46)
Diagnoses recorded	0.04	Medications (0.29)
Lab procedures	0.03	Length of stay (0.35)
Prior emergency visits	0.03	Prior outpatient visits (0.15)
Prior outpatient visits	0.02	Prior emergency visits (0.15)
Other procedures	0.00	Medications (0.37)

Weak correlations help explain why the model's AUC sits at 0.65. No single routine measure separates patients who return from those who do not.

8. Risk model

A logistic regression was trained on 70% of patients and tested on the other 30% (20,997 patients). Rare admission categories were grouped so every level has enough patients to estimate. Each patient appears once, so no patient sits in both samples.

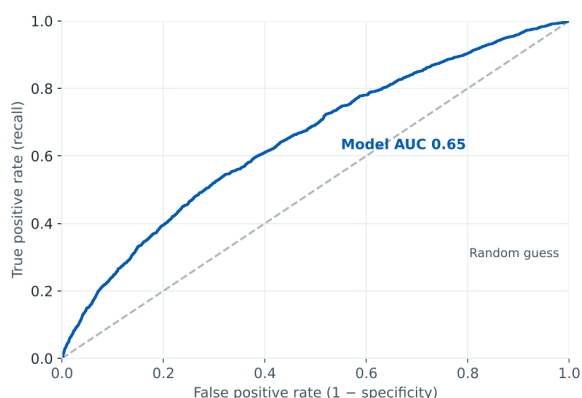
Evaluation measure	Result	What it shows
AUC-ROC, hold-out test set	0.65 (95% CI 0.64 to 0.66)	Ranks risk clearly better than chance (0.50)
AUC-ROC, 5-fold cross-validation	0.64 (SD 0.01)	Performance holds across different samples

Evaluation measure	Result	What it shows
PR-AUC	0.16, against 0.09 for a random model	Nearly doubles precision over random selection
Brier score	0.080, against 0.082 for a no-skill model	Small gain in probability accuracy
Mean predicted risk	9.0%, against 9.0% observed	Predicted risk matches the real rate overall

These results sit in the typical range for models built only on administrative data. The model is useful for ranking patients, not for individual clinical decisions.

The model separates risk better than chance (AUC 0.65)

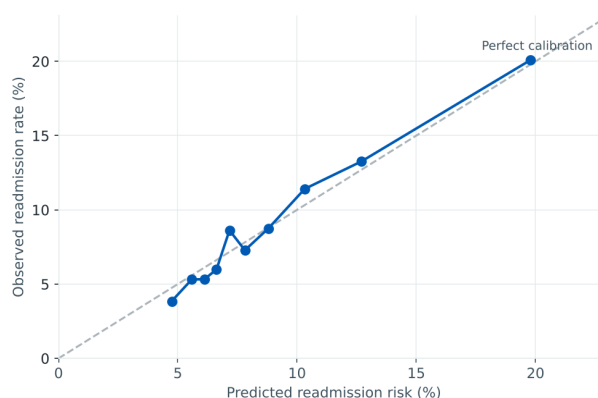
ROC curve for the logistic regression on the 30% hold-out test set



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 20,997 test patients. Analysis: C. Nkwopara.

Predicted risk tracks observed readmission reasonably closely

Calibration across 10 equal-sized risk groups. Brier score 0.080, no-skill model 0.082



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 20,997 hold-out test patients. Analysis: C. Nkwopara.

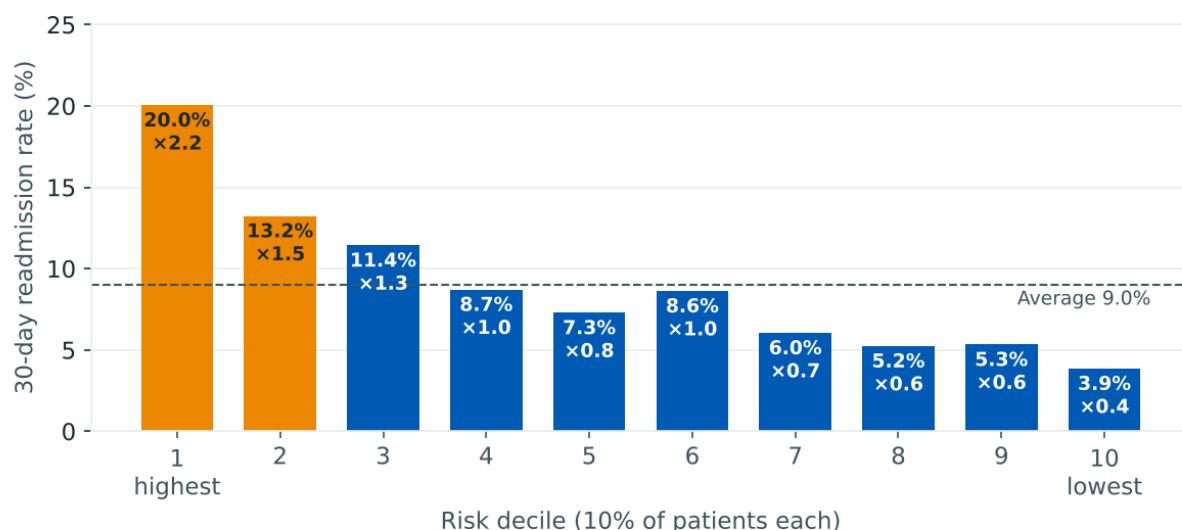
Predicted risk tracks observed readmission reasonably closely across the ten risk groups, supporting use of the scores for risk stratification and approximate probability interpretation.

8.1 Follow-up capacity scenarios

Ranking patients by risk score concentrates readmissions. The highest-risk 10% readmit at 20.1%, 2.2 times the average. The lowest-risk 10% readmit at 3.9%. The lower panel shows the share of all readmissions each follow-up capacity level would reach.

The top 20% of risk scores hold 37% of readmissions

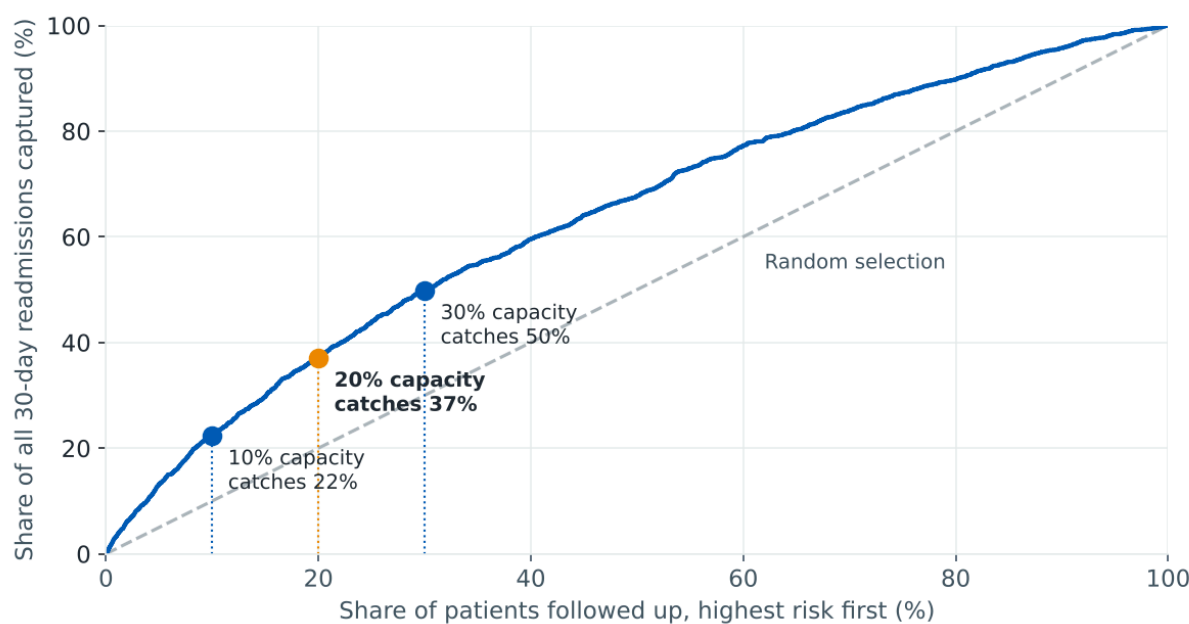
Actual readmission rate in each risk decile, with lift against the average



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 20,997 test patients. Analysis: C. Nkwopara.

Follow-up capacity sets how many readmissions the programme can reach

Cumulative gains from ranking patients by risk score. The threshold is a planning choice



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 20,997 hold-out test patients. Analysis: C. Nkwopara.

Actual follow-up capacity is unavailable in this simulated case, so no single threshold is fixed. Each capacity level below is a planning scenario. In practice, the threshold should be set against available follow-up capacity, intervention cost and the recall the service wants.

Capacity	Patients followed up per 1,000 discharges	Readmissions captured (recall)	Precision	Lift
10%	100	22%	20.0%	2.23
20%	200	37%	16.6%	1.85

Capacity	Patients followed up per 1,000 discharges	Readmissions captured (recall)	Precision	Lift
30%	300	50%	14.9%	1.66
40%	400	60%	13.4%	1.49

The 20% scenario is the worked example below and in section 9. At that level, one in six flagged patients is readmitted (precision 16.6%), against one in eleven overall.

Flagging 20% of patients catches 37% of readmissions

Precision 16.6%, recall 37.1%, F1 0.23. Rows: actual. Columns: predicted

Not readmitted	15,610 True negative	3,501 False positive
Readmitted	1,187 False negative	699 True positive
	Not flagged	Flagged for follow-up

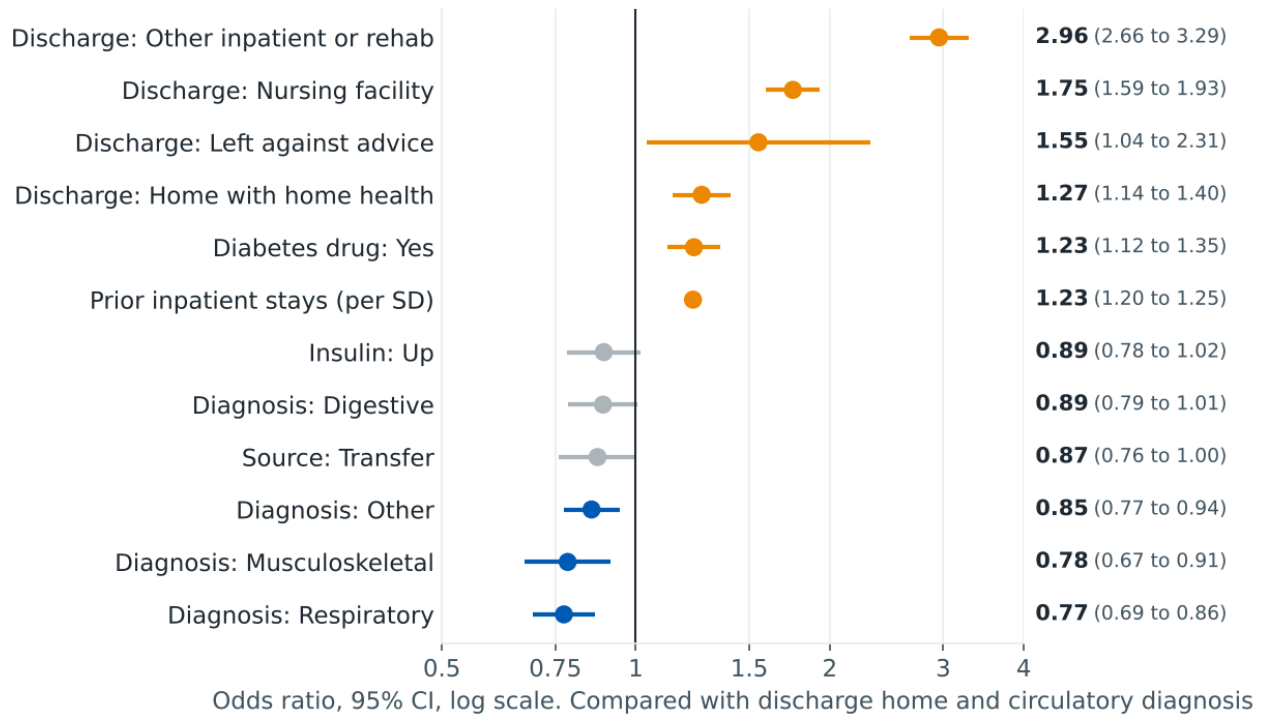
Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 20,997 test patients. Analysis: C. Nkwopara.

8.2 Risk factors

Adjusted for everything else, discharge destination dominates. Patients sent to other inpatient care or rehab have about three times the odds of readmission of those sent home (odds ratio 2.96, 95% CI 2.66 to 3.29). Nursing facility discharge raises the odds by 75%. Each standard deviation of prior inpatient stays raises them by 23%.

Discharge to other care and prior stays drive readmission risk

Adjusted for all other factors. Orange raises risk, blue lowers it, grey is not significant



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. Model trained on 48,990 patients. Analysis: C. Nkwopara.

8.3 Subgroup performance

At the 20% scenario, the model performs differently across groups. African American and Caucasian patients readmit at the same rate (9.1%), but African American patients are flagged less often (14.8% against 21.5%), and the model catches fewer of their readmissions (recall 32% against 38%). Patients aged 80 and over are flagged at 38%, nearly four times the rate for under-60s.

Group	Patients	Readmission rate	Flagged	Recall	Precision	False positive rate
Female	11,247	8.9%	21.4%	39.5%	16.5%	19.6%
Male	9,750	9.0%	18.4%	34.3%	16.8%	16.8%
African American	3,780	9.1%	14.8%	32.4%	19.8%	13.1%
Caucasian	15,718	9.1%	21.5%	38.2%	16.2%	19.9%
Other or unknown race	1,499	7.3%	17.1%	37.3%	16.0%	15.5%
Under 60	7,055	7.0%	9.7%	27.4%	19.8%	8.3%
60 to 79	10,006	9.8%	20.4%	36.0%	17.2%	18.7%
80 and over	3,936	10.6%	37.6%	50.8%	14.3%	36.0%

The age gap partly reflects real risk, since older patients readmit more often. The race gap does not, because readmission rates match. Race was not included as a model input. The observed performance gap may arise through other model inputs, underlying population differences, model specification or other unobserved factors. The cause requires investigation before any deployment (section 10).

8.4 Fairness root cause

A first investigation points to two causes: case mix, and risk the model does not capture for one age group.

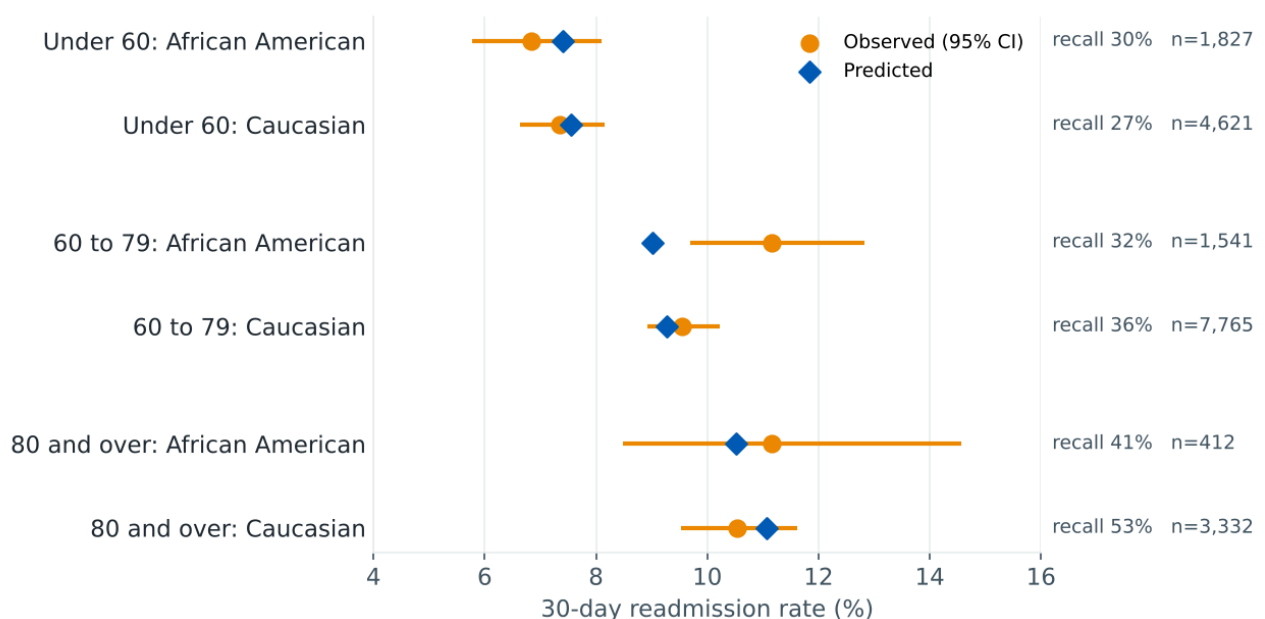
Case mix explains part of the flagging gap. In the test set, African American patients are younger and less often discharged to nursing or rehab care. Both inputs lower the risk score.

Measure	African American	Caucasian
Patients	3,780	15,718
Observed readmission	9.1%	9.1%
Mean predicted risk	8.4%	9.1%
Flagged at the 20% scenario	14.8%	21.5%
Mean age (band midpoint)	60	67
Aged under 60	48.3%	29.4%
Any prior inpatient stay	9.9%	12.1%
Discharged to nursing or rehab care	14.5%	20.8%

Case mix does not explain all of it. Overall, the model predicts 8.4% risk for African American patients against 9.1% observed. The gap sits in patients aged 60 to 79: observed readmission is 11.2% (95% CI 9.7% to 12.8%), but the model predicts 9.0%, below the whole interval. Recall in that group is 32%, against 36% for Caucasian patients. Under 60, recall is similar (30% against 27%). Results for patients aged 80 and over are too uncertain to interpret, with only 412 African American patients.

The model under-predicts risk for African American patients aged 60 to 79

Mean predicted risk against observed readmission, by race within age group, at the 20% scenario



Source: UCI Diabetes 130-US Hospitals, 1999 to 2008. 20,997 hold-out test patients. Analysis: C. Nkwopara.

One possible explanation is risk factors the dataset does not record, such as social circumstances or access to care after discharge. This analysis does not establish cause. Next steps: recalibrate the

score within subgroups, test whether added risk factors or age interaction terms close the gap, and recheck after each change.

Code: 04_analysis.py fits and tests the model. 05_model_evaluation.py produces the calibration, capacity, subgroup and business case outputs. 06_further_analysis.py produces the HbA1c variation, correlation and fairness root-cause outputs.

9. Recommendations and business case

Meridian should target follow-up by risk score and discharge route, not apply it to every diabetic patient.

Priority	Action	Evidence	Owner
1	Pilot a capacity-matched follow-up programme: set the threshold to available follow-up capacity and contact the highest-risk group within 7 days, using a locally validated and governed risk score	At the 20% scenario, the flagged group holds 37% of readmissions and readmits at 16.6% against 9.0% overall	Discharge planning leads
2	Set a handover protocol with rehab units and nursing facilities: medication list, care plan and a named contact	Odds ratio 2.96 for rehab or other inpatient care, 1.75 for nursing facility	Director of Clinical Operations
3	Refer patients with 2 or more inpatient stays in the past year to case management	Readmission 18.5% at 2 stays, 29.3% at 4 or more	Diabetes clinical lead
4	Review HbA1c testing practice against applicable clinical guidance, and check whether eligible patients are tested consistently	Only 18.4% tested, falling to about 10% for elective and musculoskeletal admissions. Half of tested patients scored above 8%. The link to readmission is negligible (V 0.010), so this is a care quality review, not a readmission fix	Diabetes clinical lead
5	Recalibrate the score by subgroup and test added risk factors, then recheck fairness before rollout	Recall 32% against 38% for Caucasian patients at the same 9.1% readmission rate. Part is case mix, but the model also under-predicts risk for African American patients aged 60 to 79 (9.0% predicted, 11.2% observed)	Analytics team
6	Report median length of stay, and track readmission monthly on an SPC chart	Stay data is right-skewed. Monthly tracking needs admission dates	Analytics team

Business case

Meridian is fictional and has no real cost data, so the business case uses hypothetical inputs. It shows how the decision would be modelled, not Meridian's actual return. All figures are per 1,000 diabetic discharges at the 20% capacity scenario: 200 patients followed up, with about 33 future readmissions in that group.

Assumption or result	Conservative	Base	Optimistic
Follow-up cost per patient (hypothetical)	\$150	\$100	\$75
Share of flagged readmissions prevented (hypothetical)	5%	10%	20%

Assumption or result	Conservative	Base	Optimistic
Cost per readmission (hypothetical)	\$10,000	\$12,500	\$15,000
Readmissions prevented	1.7	3.3	6.7
Programme cost	\$30,000	\$20,000	\$15,000
Gross saving	\$16,600	\$41,600	\$99,900
Net benefit	-\$13,400	\$21,600	\$84,900
Cost per prevented readmission	\$18,000	\$6,000	\$2,300
Break-even prevention rate	9.0%	4.8%	3.0%

In the base case, the programme breaks even if follow-up prevents about 5% of flagged readmissions. Under conservative assumptions it loses money unless prevention reaches 9%. Under base assumptions, net benefit rises with capacity up to about 40% (about \$26,900 per 1,000 discharges), then levels off.

The finance director should replace these inputs with real costs, and take the prevention rate from published evidence or a pilot.

10. Limitations and governance

Limitation	Effect on findings	Next step
Data from 1999 to 2008	Care practice has changed, so rates may not hold today	Rerun on current data
No admission dates	No trend, seasonality or SPC analysis	Request dated extracts
Readmissions only counted within the 130 hospitals	Returns to other hospitals are missed, so the rate is understated	Link to regional or national records
No social factors such as living alone, deprivation or carer support	Model misses known drivers, which caps the AUC	Add social care and deprivation data
Model AUC 0.65	Useful for ranking, not for individual decisions	Test gradient boosting, and add clinician judgement to the flag
Observational data	Links are associations, not proof of cause	Pilot the follow-up programme with a comparison group
Race recorded in the data	The model flags African American patients less often at the same readmission rate (section 8.3)	Recalibrate by subgroup and test added risk factors before rollout (section 8.4)
One index stay per patient	Repeat-stay patterns are not modelled	Retain longitudinal encounters and use patient-grouped validation or repeated-measures methods

Governance requirements before any live use of the risk score:

- Data protection: this portfolio analysis uses a publicly available research dataset and no identifiable NHS patient data. A live NHS deployment using patient-level information would require an appropriate lawful basis, access controls, data-minimisation measures and an assessment of

data-protection risks, including whether a DPIA is required. In a real US deployment, applicable federal and state privacy requirements would need to be assessed.

- Clinical safety: the portfolio model is an analytical prototype and is not intended for clinical use. If developed into a digital product influencing patient-level care in England, its intended use and applicability under DCB0129, DCB0160 and medical device regulation would need formal assessment before deployment. Any such tool would support clinician judgement, not replace it.
- Fairness: precision, recall and false positive rate are checked across race, sex and age groups. Sections 8.3 and 8.4 show the first check and a root-cause investigation. The gap must be resolved before rollout.
- Monitoring: AUC, precision and recall are rechecked each quarter, with retraining if performance drops.
- Transparency: staff know how the flag is produced, and each follow-up decision is recorded.

11. NHS application

The analytical approach can be adapted to an NHS trust, although the findings and risk estimates would need to be revalidated using current NHS data.

Meridian finding	NHS equivalent
Payer penalties for 30-day readmission	Emergency readmissions within 30 days are monitored through an NHS England national indicator, with its own definitions and exclusions
Discharge to rehab or nursing facility carries the highest risk	Discharge to intermediate care, community hospitals or care homes
7-day follow-up for the capacity-defined high-risk group	Virtual ward review, community nursing visit or GP practice follow-up
Case management for frequent attenders	High-intensity user services and integrated care teams
HbA1c testing gap	Routine HbA1c monitoring recommended in NICE diabetes guidance
Analytics stack: Python, SQL, Power BI	Standard NHS analytics stack, with SPC charts for performance reporting

12. Technical deliverables

The project ships as code, tables, charts and a dashboard file.

Deliverable	Contents
run_all.py	Runs the seven steps in order with one command
src/01_clean_data.py	Cleans the raw data and writes readmissions_clean.csv and cleaning_log.csv
src/02_visualise_cleaning.py	Charts 01 to 03: data quality and cleaning
src/03_descriptive_stats.py	Charts 04 to 12, descriptive_summary.csv, readmission_by_group.csv
src/04_analysis.py	Charts 13 to 17, FDR-adjusted tests, cross-validated model, odds ratios with confidence intervals

Deliverable	Contents
src/05_model_evaluation.py	Charts 18 and 19, calibration, capacity scenarios, subgroup performance and business case tables
src/06_further_analysis.py	Charts 20 and 21, HbA1c testing by group with z-scores, Spearman correlations, fairness root cause
src/07_doc_figures.py	Pairs related charts into the combined figures used in this report
src/style.py	Shared folder paths and NHS chart style
outputs/powerbi/readmissions_scored.csv	Risk scores for 20,997 test patients, generated locally by run_all.py for a Power BI dashboard. Not published in the public repository
This case study	Business context to recommendations

13. Data source, licence and use

This portfolio case study uses the [Diabetes 130-US Hospitals for Years 1999–2008](#) dataset by John Clore, Krzysztof Cios, Jon DeShazo and Beata Strack, UCI Machine Learning Repository, DOI 10.24432/C5230J, licensed under [CC BY 4.0](#). The source data were cleaned, recoded, grouped and transformed for this analysis. UCI and the dataset creators do not endorse this case study or the fictional Meridian Health Network scenario.

Citation: Clore, J., Cios, K., DeShazo, J., and Strack, B. (2014). Diabetes 130-US Hospitals for Years 1999–2008 [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5230J>

Meridian Health Network and its stakeholder exercise are fictional. No real stakeholder interviews were conducted. Financial assumptions are illustrative.

The predictive model is a retrospective analytical prototype created for portfolio demonstration using historical public data. It has not been clinically validated, certified or approved for patient-care decisions. Any real healthcare deployment would require current local data, clinical validation, information-governance review, equality and fairness assessment, clinical-safety assessment and consideration of applicable medical device requirements.